

Harsha S. Bhat

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EDUCATION

- 2021 **H.D.R., Supershear Earthquakes** — École Normale Supérieure, France[†]
- 2007 **Ph.D., Mechanical Sciences** — Harvard University, USA *
- 2002 **M.S., Engineering Sciences** — Harvard University, USA
- 2001 **B.E., Civil Engineering** — NITK, India

[†] *Habilitation à Diriger des Recherches* * *Supervised by Prof. James R. Rice & Dr. Renata Dmowska*

CURRENT POSITIONS

- 2024 **Directeur de recherche CNRS** — École Normale Supérieure, France
- 2026 **Professeur attaché** — École Normale Supérieure, France

PAST POSITIONS

- 2022 – 2026 Professeur chargé de cours — École Polytechnique, France
- 2016 – 2024 Chargé de recherche CNRS — École Normale Supérieure, France
- 2021 – 2023 Visiting Professor — NISER, India
- 2012 – 2016 Chargé de recherche CNRS — Institut de Physique du Globe de Paris, France
- 2010 – 2011 Assistant Professor (Research) — University of Southern California, USA
- 2007 – 2010 Post-Doctoral Fellow — University of Southern California, USA
- 2007 – 2010 Visitor in Aeronautics — California Institute of Technology, USA
- 2007 – 2007 Post-Doctoral Fellow — Harvard University, USA
- 2001 – 2007 Graduate Research Associate — Harvard University, USA

FUNDING & GRANTS

* → *waiting for results* gray → *unsuccessful*

- 2026* Agence Nationale de la Recherche – PRCI, France-HongKong coPI
“REFRESH: Revisiting Seismic Hazard and Fault Mechanics in Intraplate Regions under Climatic Stresses”
- 2026* Partenariat Hubert Curien PROCOPE, France–Germany coPI
“CASPER: Climate-Aware Seismic Hazard from Physics-Based Earthquake-Cycle Models”
- 2026 Domaine d’Intérêt Majeur MaTerRE, Île-de-France coPI
“GEO-PROBE: GPU-Accelerated Multi-Physics Spectral Boundary-Element Modeling of Induced Seismicity during Underground Gas Storage”
- 2024 Laboratoire de Recherche Conventionné (CEA), France coPI
“CLIMSEISM: Climate-Triggered Intraplate Seismicity — Detection and Mechanics”
- 2023 Agence Nationale de la Recherche (AAPG 2022), France coPI
“SMEC: Smectite Dehydration — a Key to Fault Mechanics”
- 2023 Partenariat Hubert Curien PROCORE, France–HongKong coPI
“Earthquake Risks along the Anninghe Fault Zone”
- 2022 Partenariat Hubert Curien Aurora, France–Norway coPI
FARM: Fault and Rupture Mechanics
- 2022 CNRS – University of Tokyo, France–Japan coPI
“SESAME: Seasonal Seismicity — Statistical Analysis and Modeling”

2021	Centre National d'Études Spatiales (CNES), France <i>On the Link between Rupture Velocity and Off-Fault Damage Produced in Earthquakes</i>	coPI
2020	Endeavour Fund (MBIE), New Zealand <i>Can Earthquake Simulators Forecast Seismic Hazard after the Next Great Alpine Fault Earthquake?</i>	coPI
2019	European Research Council, Consolidator Grant, EU <i>"PERSISMO: Predicting Energy Release in Fault Systems — Integrating Simulations, Machine Learning and Observations"</i>	PI
2019	Institut National des Sciences de l'Univers (CNRS), TelluS-ALEAS, France <i>IRA: Inverting Rupture Velocity Using Damage and Aftershocks</i>	PI
2019	Marie Skłodowska-Curie ITN (H2020), EU <i>FRICFAULT: Friction in Faults</i>	coPI
2019	Émergence (Ville de Paris), France <i>Inelastic Deformation in the Subduction Zone</i>	coPI
2019	Laboratoire de Recherche Conventionné (CEA), France <i>"Earthquake Dynamics, Damage & Tsunamigenesis"</i>	coPI
2019	Aramont Fund (Harvard), USA <i>"Resolving Earthquake Source Mechanisms on Complex Fault Systems (fellow: K. Okubo)"</i>	coPI
2019	Scheme for Promotion of Academic and Research Collaboration (SPARC), India <i>"Novel Explorations on Earthquake Nucleation within Complex Fault Networks"</i>	coPI
2018	PRESTIGE Grant, France <i>"Seismic Hazard on Thrust Faults Using Physics-Based Models and Surface-Data Analysis (fellow: L. Bruhat)"</i>	host
2018	Thomas Jefferson Fund (FACE Foundation), France–USA <i>Earthquakes in Complex Fault Networks</i>	coPI
2018	École Normale Supérieure (Actions Incitatives), France <i>"Earthquake Mechanics"</i>	PI
2017	Fondation Simone et Cino Del Duca (Académie des sciences), France <i>Dynamics of Fault Networks: Physics-Based Models</i>	PI
2017	Institut National des Sciences de l'Univers (CNRS), mi-lourds, France <i>Observational and Computational Earthquake Cycles</i>	PI
2017	Thomas Jefferson Fund (FACE Foundation), France–USA <i>Earthquakes in Complex Fault Networks</i>	coPI
2016	Agence Nationale de la Recherche, France <i>LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves</i>	PI
2016	Institut National des Sciences de l'Univers (CNRS), TelluS-ALEAS, France <i>"Spontaneous Off-Fault Damage Due to Earthquakes and the Associated Acceleration Spectra"</i>	PI
2016	Institut de Physique du Globe de Paris (Mobilité), France <i>"FDEM Rupture Modeling at Los Alamos (PhD mobility: K. Okubo)"</i>	coPI
2015	Agence Nationale de la Recherche, France <i>LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves</i>	coPI
2015	Émergence (Ville de Paris), France <i>DEED: Deep Earthquake Dynamics</i>	coPI
2015	Natural Environment Research Council (Fellowship), UK <i>Seismic and Aseismic Slip: Properties and Constitutive Behavior of Faults</i>	coPI
2015	Université Sorbonne Paris Cité, France <i>"Earthquake Ruptures on Complex Fault Systems and Complex Media (PhD fellowship K. Okubo)"</i>	PI
2014	European Research Council, Starting Grant (H2020), EU <i>UNITERRA: Unifying On- and Off-Fault Earthquake Models</i>	PI
2014	Émergence (Ville de Paris), France <i>Quantifying the Seismic Hazard Associated with Waste-Water Injection</i>	PI
2014	Natural Environment Research Council, UK <i>Earthquake Fracture Damage and Feedbacks in the Seismogenic Zone</i>	coPI
2014	Agence Nationale de la Recherche, France <i>LIVOGERM: Lithospheric Volcanism and Geothermal Energy at Major Strike-Slip Fault Intersections</i>	coPI

2014	Université Sorbonne Paris Cité, France LABSEISM: Laboratory Investigation of Earthquakes Triggered by Seismic Waves	coPI
2013	European Research Council, Starting Grant, EU SEISMECH: Towards the Next Generation of Mechanical Models of Earthquake Source Physics	PI
2013	Émergence (Ville de Paris), France GEOSEISMECH: Integrating Mechanical Models with Geological, Experimental and Seismological Observations	PI
2013	AXA Research Fund, France “Seismic Hazard of the Shallow Portion of Megathrusts (postdoc: N. Cubas)”	host
2013	Southern California Earthquake Center (SCEC), USA “Dynamic Formation of Fault-Zone Rocks and Their Influence on Rupture Propagation”	PI
2012	Agence Nationale de la Recherche (Chaires d’Excellence), France GEOSEISMECH: Geological and Seismological Mechanics	PI
2012	Agence Nationale de la Recherche (Blanc, SIMI 6), France GeoSMEC: Geometry of Strike-Slip Faults through Multiple Earthquake Cycles	PI
2011	Émergence (Université Pierre et Marie Curie), France From Damage to Rupture of Quasi-Brittle Materials: Application to Seismic Faults	coPI
2011	National Science Foundation (instrumentation), USA “Optical Interferometer Assembly for Time-Resolved Pointwise Stress & Velocity Measurements in Laboratory Earthquakes”	coPI
2011	Southern California Earthquake Center (SCEC), USA “The Effect of Off-Fault Damage on the Propagation of an Earthquake Rupture”	coPI
2010	National Earthquake Hazards Reduction Program (USGS), USA “Geometric Heterogeneities in Rupture Propagation for the Kunlun–Kunlun Pass–Xidatan Fault System”	PI
2010	National Science Foundation, USA “A Rate-Dependent Micromechanical Damage Mechanics for Simulating Dynamic Earthquake Ruptures”	PI
2010	National Nuclear Security Administration / U.S. Air Force, USA “Generation of High-Frequency P and S Energy by Rock Fracture During a Buried Explosion”	coPI
2008	National Science Foundation, USA Laboratory Earthquakes: Characterization of Ground Motion and Stress States in Complex Rupture Scenarios Using High-Resolution Optical Diagnostics	coPI
2008	National Science Foundation, USA “Micromechanics-Based Modeling of Dynamic Earthquake Rupture in a Structurally Complex Fault Zone”	coPI

HONORS & AWARDS

2018	Grand Prix Michel Gouilloud Schlumberger — French Academy of Sciences
2018	CNRS Award for Doctoral Supervision and Research
2003 – 2006	Harvard University Certificate of Distinction in Teaching (2003, 2004, 2006)

STUDENTS

Postdoctoral Associates

2024 – 2026	Ankit Gupta (India)	
2025 – 2027	Suli Yao (China)	
2022 – 2025	Navid Kheirdast (Iran)	Post-Doctoral Fellow, ISTeP
2021 – 2023	Michelle Almakari (France)	
2021 – 2023	Carlos D. Villafuerte (Mexico)	Asst. Prof., UNAM
2019 – 2022	Ekeabino Momoh (Nigeria)	AXA Postdoc Fellow
2018 – 2021	Lucile Bruhat (France)	Nat. Catastrophe Risk Analyst, AXA
2019 – 2019	Lisa Gordeliy (Russia)	Senior Engineer, ResFrac

2014 – 2016 Marion Y. Thomas (France) CNRS Scientist, Univ. de Rennes

PhD Students

2024 – 2027	Bharath Shanmugasundaram (India)	
2024 – 2027	Yishuo Zhou (China)	
2023 – 2026	Thomas Melkior (France)	
2023 – 2027	Caiyuan Fan (China)	
2021 – 2024	Jinhui Cheng (China)	Postdoc, Caltech
2020 – 2023	Augustin Thomas (France)	Research Scientist, BRGM
2020 – 2023	Joseph Flores Cuba (Peru)	Data Scientist, PowerBI
2018 – 2021	Claudia Hulbert (France)	CEO, Geolabe
2017 – 2020	Samson Marty (France)	Postdoc, Caltech
2014 – 2019	Marshall A. Martinez (USA)	Engineer, Joby Aviation
2015 – 2018	Kurama Okubo (Japan)	Research Scientist, NIED, Japan
2014 – 2017	Pierre Romanet (France)	Research Scientist, Cerema, France
2010 – 2015	Vahe Gabuchian (USA)	Research Scientist, Caltech
2011 – 2014	François X. Passelègue (France)	CNRS Scientist, GeoAzur, Nice
2008 – 2013	Jonathan Mihaly (USA)	Jet Propulsion Laboratory
2007 – 2012	Michael Mello (USA)	Teaching Professor, Caltech

Undergraduate & Masters Interns

2022	Valentin Marnat	2018	Nicolas Mercury	2013	Thibaut Perol
2021	Roxane Ferry	2018	Luc Illien	2012	Lucile Bruhat
2020	Jinhui Cheng	2017	Phillipe Danre	2004	Marion Olives
2019	Roxane Ferry	2015	Eleni Kolokytha	2003	Sonia Fliss
2019	Phillipe Danre	2015	Victor Barolle		
2019	Hugo Lestrelin	2014	Kurama Okubo		

TEACHING

Current — École Normale Supérieure

Earthquake Source Mechanics | Numerical Methods in Geosciences | Thermodynamics

Past — Harvard, Caltech, IPGP, ENS, École Polytechnique (with various colleagues)

Mécanique des Milieux Continus | Active Faults: Geometry | Seismic Ruptures and Scaling Laws | Introduction to Rock Physics | Mathematical Methods in the Sciences | Environmental Risks and Disasters | Ordinary and Partial Differential Equations | Complex and Fourier Analysis | Computational Solid and Structural Mechanics | Solid Mechanics | Introduction to the Mechanics of Solids | Mechanics of Fracture | Advanced Geomechanics | Mécanique de la Fracturation | Mécanique de la Rupture | Mécanique des Matériaux et des Structures

ORGANIZATION OF SCIENTIFIC MEETINGS

Jan 2026	Earthquake Source: Mechanics, Seismology and Geology Workshop — NISER, India
2023, 2024	Across the Time Scales, from Earthquakes to Earthquake Cycle — EGU
Jun 2019	Coupled Processes in Fracture Propagation in Geo-Materials — CISM Advanced School, Udine
Apr 2015	Multiscale Modeling of Fragmentation & Damage in Fault Zones — SSA
Dec 2014	Fault Zone Properties & Processes During Dynamic Ruptures — AGU

RESPONSIBILITIES & SERVICE

2018 – 2024 Team Leader, Faults & Earthquakes Group — ENS
2018 – 2019 Co-organizer, Internal Seminar — ENS
2025 Member, HCERES Committee for ISTerre — Univ. Grenoble
2024 – 2025 Editor, JTCAM

REVIEWING ACTIVITIES

American Geophysical Union | Seismological Society of America | Science | Nature | Nature Communications | Nature Geoscience | Science Advances | Proceedings of the National Academy of Sciences | Journal of the Mechanics and Physics of Solids | European Journal of Mechanics – A/Solids | International Journal of Fracture | Earth and Planetary Science Letters | Geophysical Research Letters | Journal of Structural Geology | Geology | Geophysical Journal International | Journal of Applied Mechanics | Geological Society of America | National Science Foundation | European Research Council

BOOKS

BOOK CHAPTERS

PUBLICATIONS

In Preparation

Under Review

Published